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5 June 1985
SUPERSEDING
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21 February 1978

NAVAL TRAINING EQUIPMENT CENTER
ORLANDO, FLORIDA

SPECIFICATION FOR

TRAINING AID, PLOTTING BOARD

DEVICE 1GG20

This specification is approved for use within Naval Training Equipment Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Training Aid, Plotting Board for aerial navigation with computer MARK 8B attached.

1.1.1 This item has been designated by the Naval Training Equipment Center as Device 1GG20 for Navy procurement (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or requests for proposals, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-794	-	Parts and Equipment; procedure for packaging and packing of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Training Equipment Center, Code 244, Orlando, FL 32813-7100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials used shall be as specified in 3.1.1 through 3.1.3.

3.1.1 Metals and fasteners. Metals and fasteners shall be either corrosion resistant steel, or brass chromium plated.

3.1.2 Plastic. Plastic shall be durable and resistant to cracking, brittleness, discoloration and deformation upon aging, or when subject to the range of temperature specified in 3.4.

3.1.3 Hardboard. Hardboard shall be tempered, smooth on both sides and moisture resistant, for a paint or lacquer finish. U.S. Gypson "Duron" or material equivalent in strength, rigidity and smoothness of surface is acceptable.

3.2 Design.

3.2.1 The training aid is designed to allow the plotting and the solution of aerial navigation problems. The base plate, with a MARK 8B computer fastened in the right hand corner, accommodates a distance, grid disc, and plotting sheet either Type A or Type B. For assembly of the plotting board, see page 9. For a view of all components of Device 1GG20, see pages 10 through 15.

3.3 Construction. Methods of construction may be chosen to suit the manufacturer's available facilities. However, the locations and tolerances of holes in the plotting sheets and buttons in the baseplate, as shown on page 15, must be met in order to assure the interchangeability of components from different sources.

3.4 Performance. The training aid shall withstand continuous classroom use and shall not be adversely affected in any manner during storage and transportation when under ambient conditions of temperature ranging from minus 20 to plus 120 degrees Fahrenheit, and of relative humidity ranging up to 95 percent. The training aid shall be so constructed that no components, assemblies or subassemblies shall bind or become undesirably free.

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3.5 Details of components. The training aid shall consist of the following major components. For graphic outline of each component see pages 9 through 15.

a. Baseplate

b. MK-8B Computer

c. Distance Grid Disc

d. Plotting Sheet Type A, three each

e. Plotting Sheet Type B, three each.

3.5.1 Baseplate. The baseplate shall be made of 3/16 tempered hardboard as defined under 3.1.3. Dimensions and location of fasteners shall correspond to page 15. The baseplate shall have a white lacquer satin finish on all surfaces. The center pin, shown on page 15, shall be made of corrosion resisting steel and shall be secured by a "TRUARC" retaining ring, Waldes Kohinoor Series 5115, or equal. Five snap fasteners, "Jot" or "Gripper" type, or equal, shall be capable of securely holding the plotting sheets. The fastener straps shall be made of suitable plastic and shall not become deformed through use or storage.

3.5.1.1 Imprint on back of baseplate. The following three groups shall be imprinted in black on the back of the Baseplate. Size of letters and arrangement shall be Contractor's choice; however, printing shall be protected from wear by suitable means (clear lacquer, fused on vinyl).

a. Device: 1GG20
Stock No: 2D-6910-534-2457
Contract No:
Manufacturer:
Year: (Year of Delivery)

b. Black and white imprint of
NAVTRAEQUIPCEN Emblem

c. NOTE: Replacement plotting sheets may be obtained by submitting a MILSTRIP document via appropriate supply channels to Naval Training Systems Center (Code 442), Orlando, FL 32813-7100

NOMENCLATURE	PART NO.	STOCK NO.
Plotting Sheet Type A	PN5528	6910-LL-C00-5200
Plotting Sheet Type B	PN5529	6910-LL-C00-5201

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3.5.2 MK-8B computer. The MK-8B Computer shall be furnished as part of Device 1GG20. The computer baseplate and the rotating disc shall be made of white vinyl, minimum .035 thick. Shape, outline and printing shall correspond to the Government-furnished sample under 3.9(b). The printing in black shall be sharp, clear, and accurate and, when assembled, shall satisfy the accuracy requirements under 4.7.1. The printing shall be protected by a fused on .005 clear vinyl. The computer shall be riveted to the baseplate of the plotting board.

3.5.3 Distance grid disc. The grid disc shall be made of white vinyl, minimum .035 thick. It shall be printed in green on both sides to correspond to the Government-furnished sample under 3.9(b). The printing shall be protected by a fused on .005 clear vinyl. The center hole shall have a sliding fit over the center post of the baseplate of the plotting board.

3.5.4 Plotting sheet type A. Three plotting sheets, Type A, shall be delivered with each Device 1GG20. The plotting sheet shall be made of transparent vinyl, minimum .020 thick. The front face shall have a matte finish which will provide a good surface for pencil marks. The back surface shall be glossy. It shall be imprinted in black on the back surface. The outline, shape and printing shall correspond to the Government-furnished sample under 3.9(b). The holes shall be at the exact center location specified on page 15 and shall be .005 inch in diameter larger than the mating pin in the plotting board. The center hole shall not have a grommet or bushing.

3.5.5 Plotting sheet type B. The plotting sheet, Type B, shall be identical to the Type A described under 3.5.4 with the exception of the imprinting. The imprinting legend on the back surface shall conform to the Type B sheet of the Government-furnished sample under 3.9(b).

3.6 Dimensions and tolerances. Dimensions and tolerances shown on page 15 shall be met. Dimensions not shown may be scaled from the Government-furnished sample under 3.9(b).

3.7 Finish. All surfaces to be painted shall be sealed, primed, and finished with a satin surface lacquer. All finished surfaces shall have no sharp edges, corners or visible imperfections.

3.8 Government-furnished property. The following property will be furnished by the Government:

a. A NAVTRAEQUIPCEN emblem will be furnished as nonreturnable sample.

b. A Device 1GG20 will be furnished to serve as a general guide and as noted under 3.5.2, 3.5.3, 3.5.4, 3.5.5, and 3.6. Upon completion of the contract, this device shall be returned to the Government.

c. In case of a difference between the furnished sample and this specification, the specification shall override.

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3.9 Workmanship.

3.9.1 All printing shall be sharp, distinct, no blurs or ghosts. The line work shall be sharp and accurate. The printing of the computer shall satisfy the test requirements under 4.7.1.

3.9.2 The training aid, when all components are assembled, shall be accurately aligned and inherently strong as a unit. All components shall be free from warpage, cracks, chips, sharp edges or other imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspections to be performed are classified as follows:

a. In-process inspection

b. Quality conformance inspection.

4.3 In-process inspection. In-process inspection shall include such visual, electrical, and mechanical examinations and testing of materials, subassemblies, parts, and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements specified herein.

4.4 Quality conformance inspection.

4.4.1 Preproduction inspection. First article approval in accordance with ASPR Part 19 is required. The inspection and tests of the first article will be conducted by the Government. The approved and accepted first article shall be the standard for the production inspection.

4.4.1.1 Preproduction inspection shall consist of all the examinations and tests specified herein.

4.4.2 Production inspection. Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction sample.

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4.5 Examinations. Examinations shall be performed to verify conformance to the following requirements:

a. Materials

b. Design

c. Construction

d. Details of components

e. Dimensions and tolerances

f. Finish

g. Nameplates and emblem

h. Workmanship.

4.6 Tests. The following tests shall be performed:

a. Operating test

b. Acceptability test.

4.7 Test methods.

4.7.1 Operating test. The training aid shall be subjected to an operating test, in normal room temperature, 68 degrees Fahrenheit (F), ±5 degrees, of not less than two hours, to insure proper functioning under these limits. The training aid shall be capable of meeting the performance requirements without alignment or adjustment, other than the necessary attachments for operation. No repairs shall be permitted during the operating tests. If repairs are required, the test shall be repeated after the necessary replacements have been made. During the operation period, not less than twelve calculations shall be performed.

4.7.2 Acceptability test. The training aid shall receive a surface inspection to assure compliance with the requirements of this specification the computer shall solve the problems of table I within the tolerance indicated:

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TABLE I. Computer tests.

Set Speed Index to	Opposite Time	Read Distance
300 knots	18 minutes	90.0 ± 0.1 miles
100 knots	5:00 hrs	500 ± 1.0 miles

Set air Speed	Opposite Air Temperature	Read Density Altitude
0 knots	+ 15 deg. C	0 ± .1 thou. ft.
300 knots	0 deg. C	34.8 ± .1 thou. ft.

Set pressure Altitude	Opposite Air Temperature	Opposite Calibrated Air Speed	Read True Air Speed
0 thou. ft.	- 60 deg. C	500 knots	369 ± 1 knots
30 thou. ft.	+ 40 deg. C	140 knots	191.9 ± 0.2 knots

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in conformance with MIL-STD-794. Level of protection shall be as specified in the contract.

5.2 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The training aid will be used as a training device for classroom instruction in aerial navigation.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Procuring activity designation (device number or other) (see 1.1.1).

c. Government-furnished samples (see 3.8).

d. Selection of applicable levels of preservation, packaging and packing (see 5.1).

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6.3 Definitions. Definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction sample. A production item furnished under production contracts for the purpose of determining compliance with requirements of the detail training device specification. It shall be representative of production sets.

6.3.2 Production unit. A unit in its final form of final production design made by production tools, jigs, fixtures and methods.

6.3.3 Training aid. A training device designed to demonstrate or illustrate a concept.

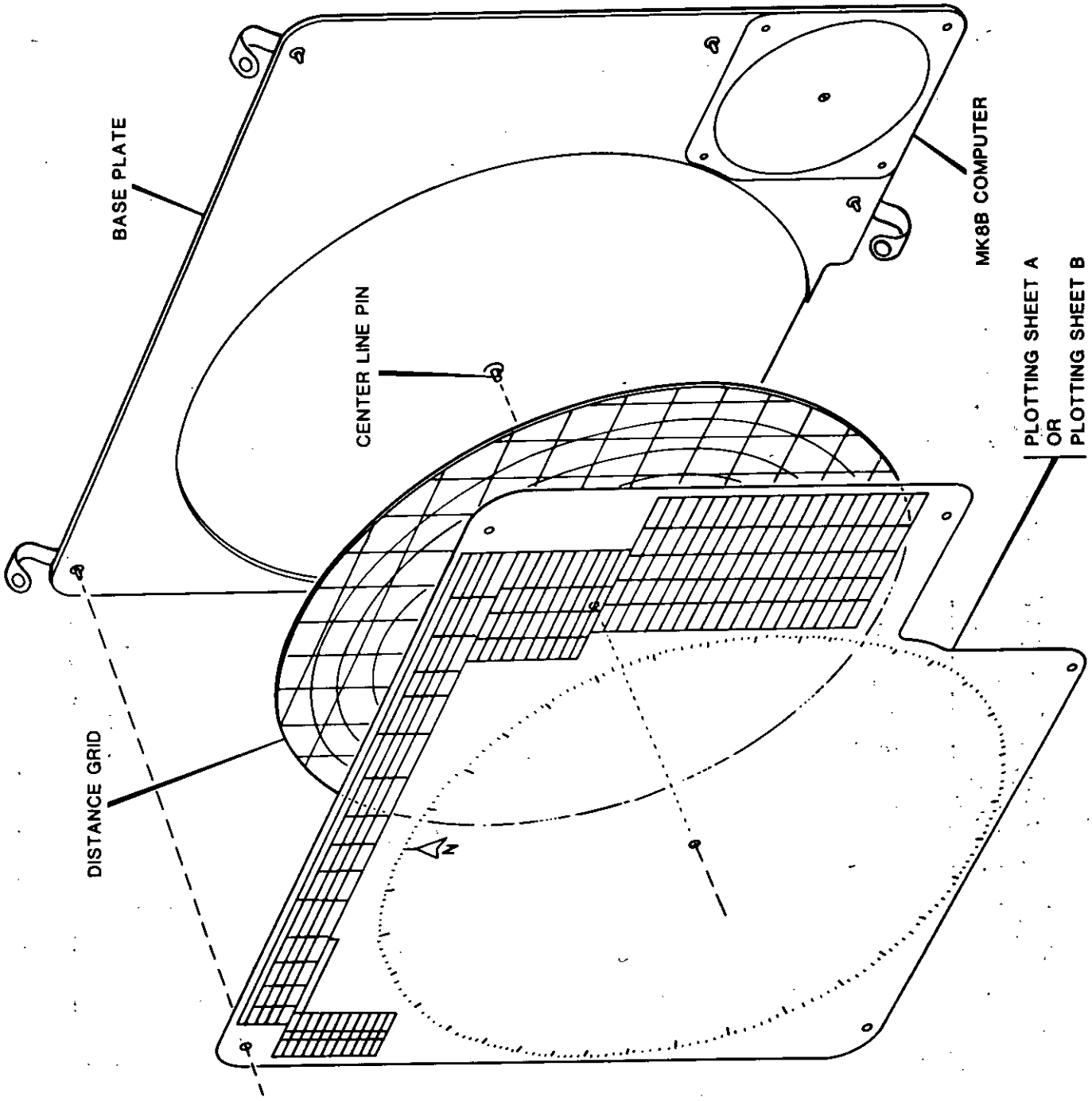
6.3.4 Changes from previous issue. Vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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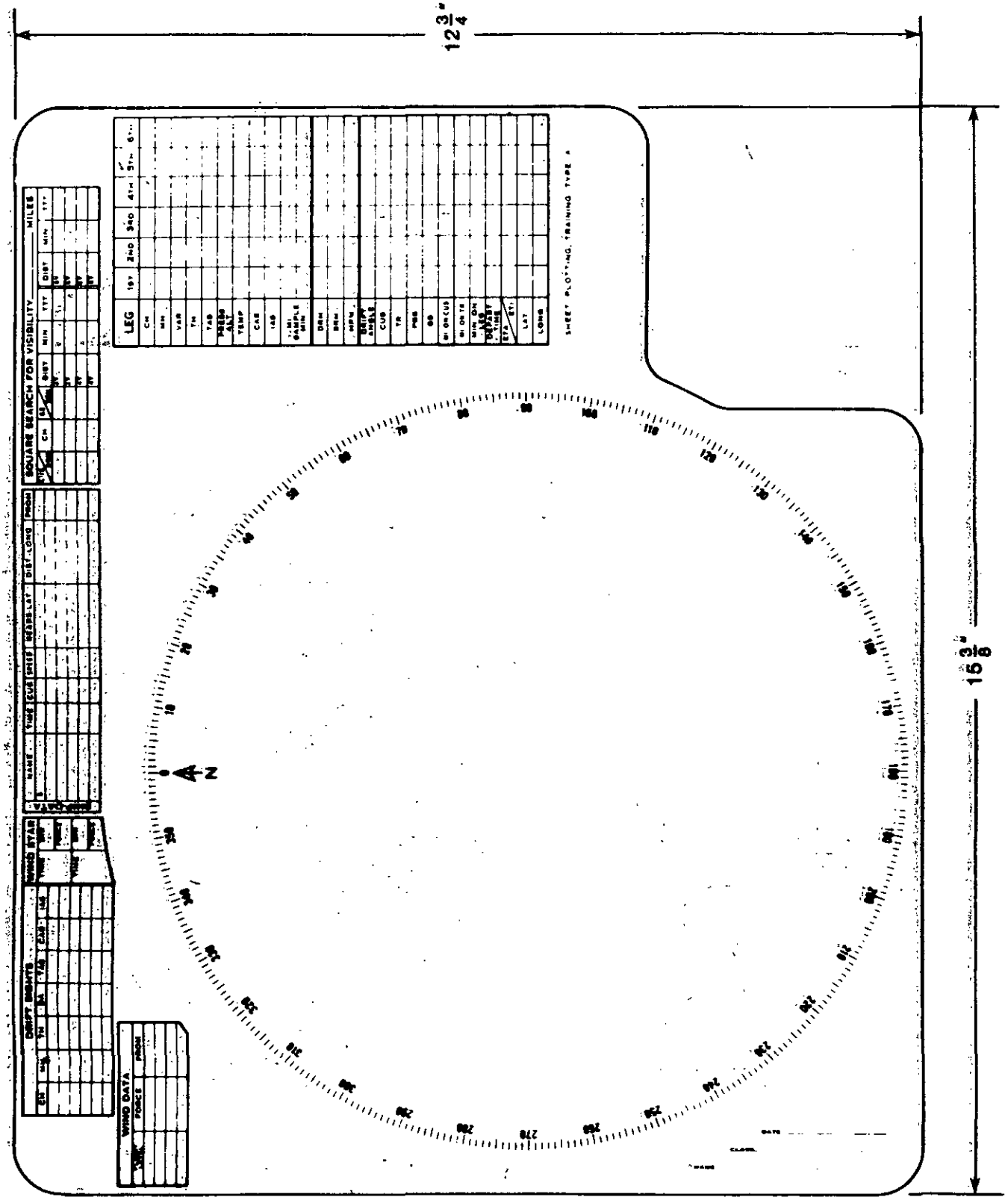
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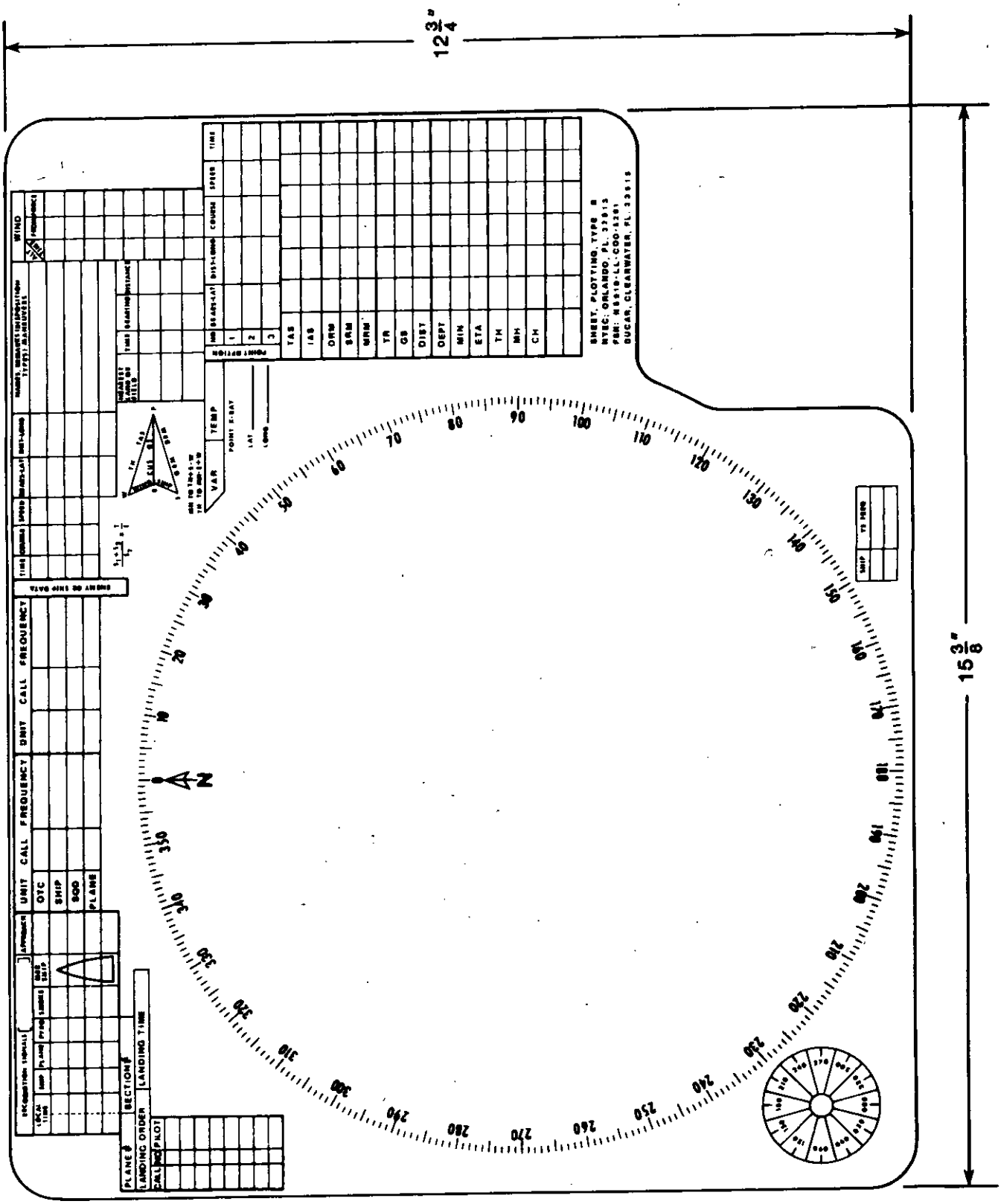
PLOTTING BOARD, DEVICE 1GG20

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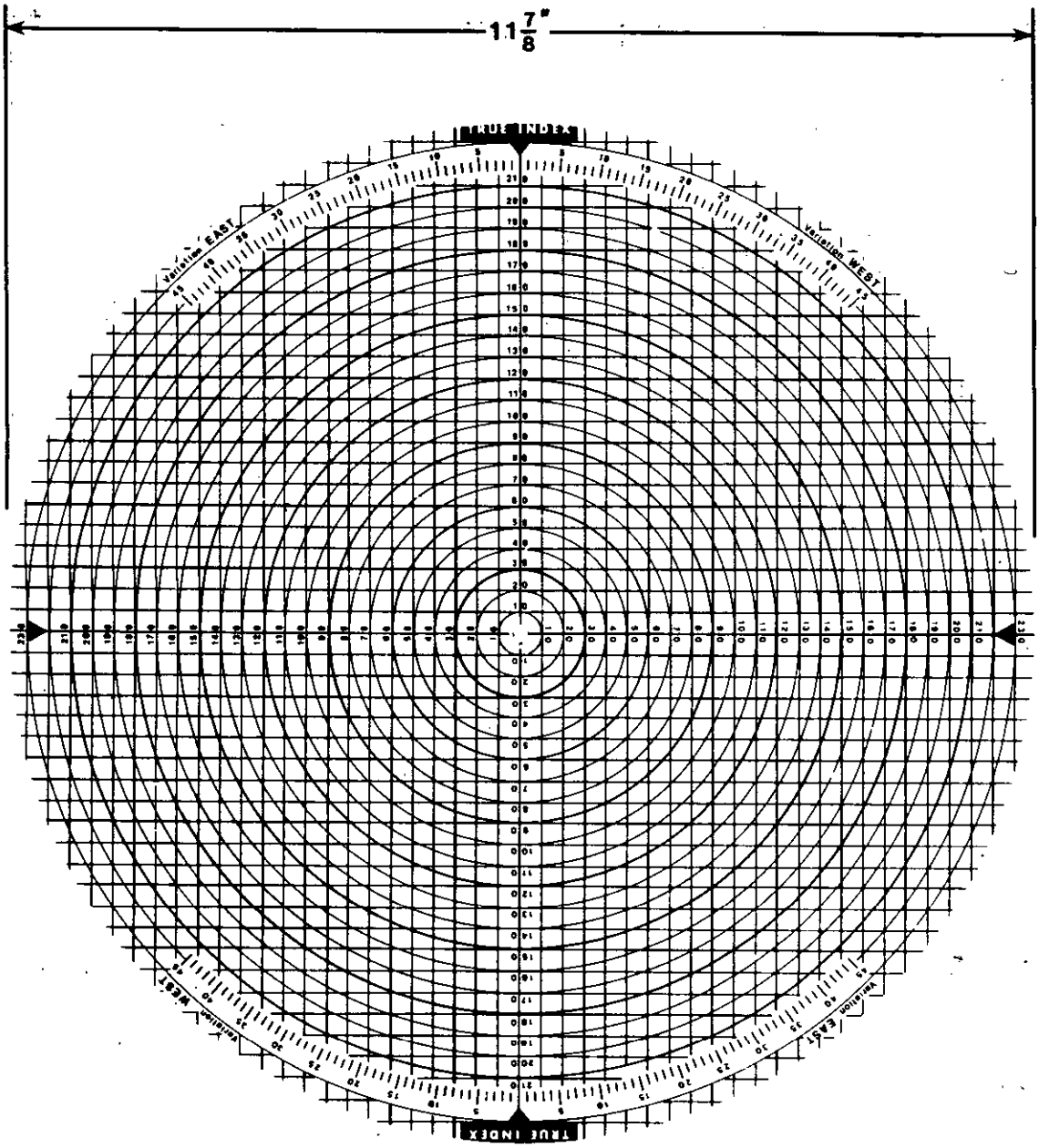
PLOTTING SHEET "A"

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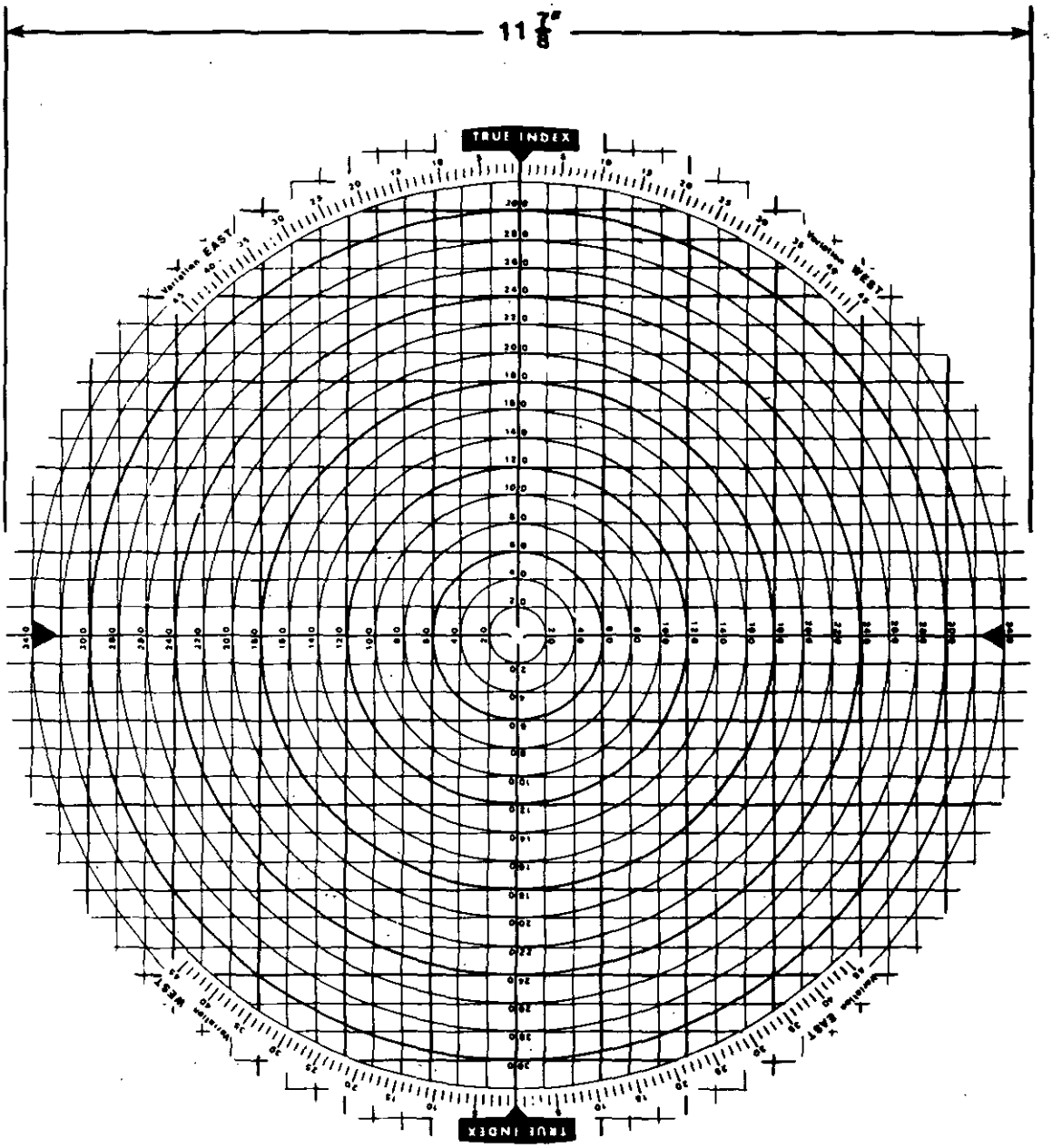
PLOTTING SHEET "B"

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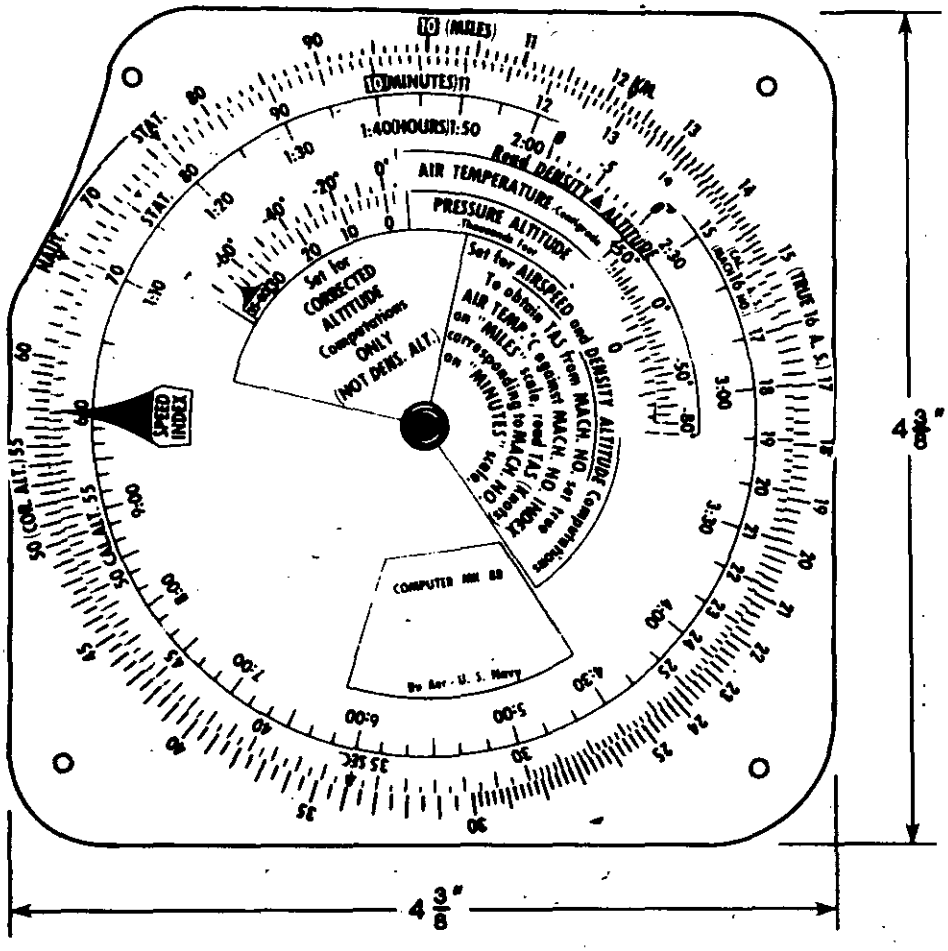
DISTANCE GRID (230 MI.)

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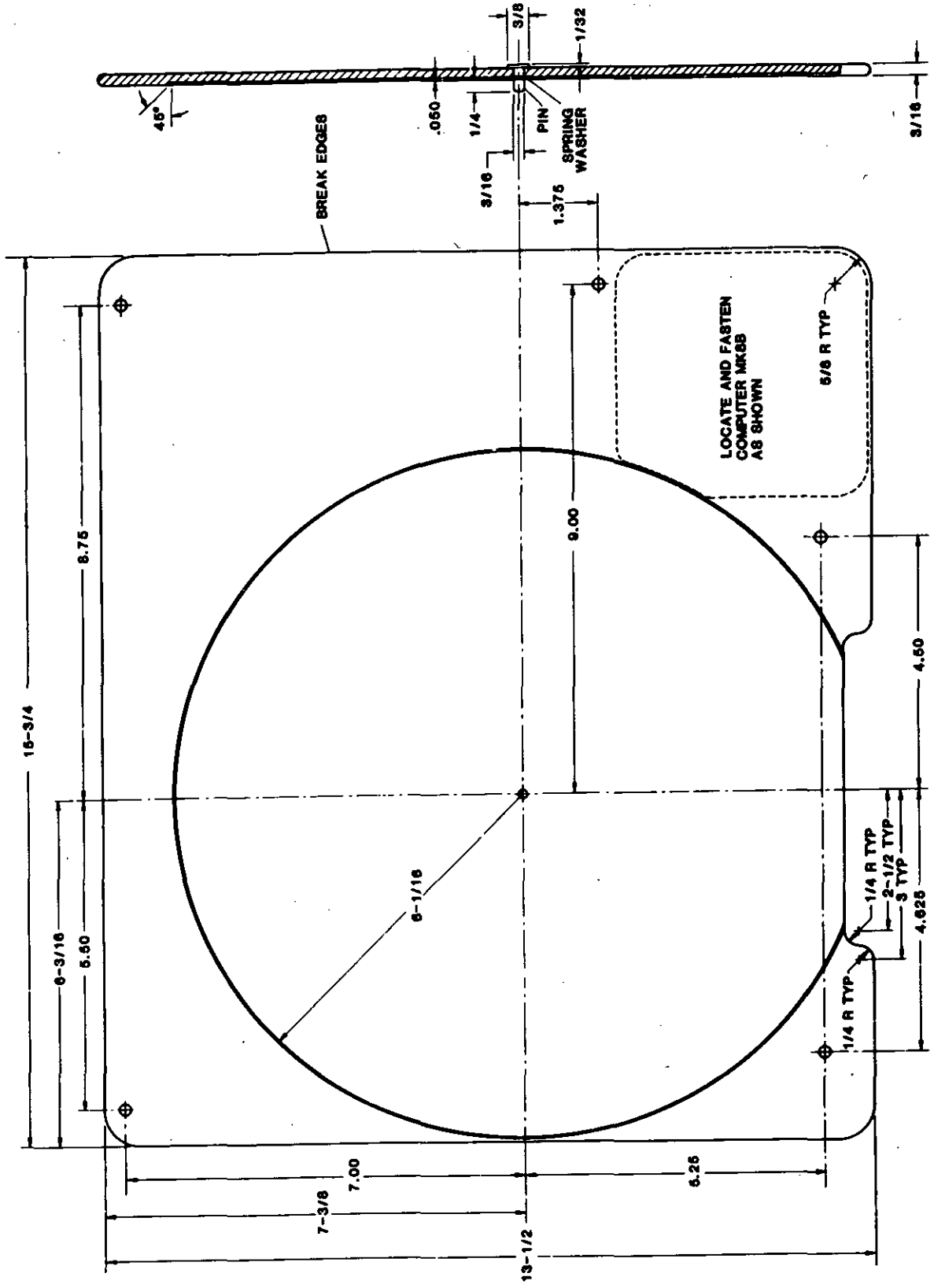
DISTANCE GRID (340 ML)

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MK8B COMPUTER

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UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
TOLERANCES: FRACTIONS ± 1/32
DECIMALS ± .003

BASE PLATE

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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